CDEQ V3.0: Self-Sustained Aerothermal Propulsion via Critical Velocity Threshold Switching 气动热摩擦 自持循环

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Abstract

This paper presents CDEQ V3.0 (Cyber-Dynamic-Equilibrium-Control Version 3.0), a self-sustained propulsion framework based on aerothermal friction heat recovery and velocity-dependent mode switching. The system uses multi-stage compressed air tanks to accelerate beyond a critical velocity threshold, at which point aerodynamic heating becomes sufficient to maintain continuous cruise thrust.

CDEQ V3.0 is strictly established on 13 universal axioms from classical physics, thermodynamics, and stability theory, plus 3 boundary axioms that ensure compliance with thermodynamic laws. The system does not constitute perpetual motion and follows all conservation laws.

This work extends the aerothermal philosophy of the Qian Xuesen trajectory and provides a complete, verifiable, and realizable architecture for long-endurance flight.

Keywords: CDEQ V3.0, self-sustained propulsion, aerothermal heating, friction heat recovery, critical velocity, GCVT

1. Introduction

The Qian Xuesen trajectory achieves supreme tactical performance through high-speed wave-riding, like a high-speed boat surging on the ocean surface.
CDEQ V3.0 pursues extreme endurance via underwater-style self-sustained cruise: lower peak speed, but nearly unlimited range.
The two represent distinct paths, yet both are rooted in the same aerodynamic and thermal foundation.

CDEQ V3.0 originates from a universal physical observation: motion generates frictional

heat, and sufficient heat can sustain motion. The system consists of two phases:

1. Tank-driven acceleration to reach critical velocity
2. Aerothermal self-sustained cruise above critical velocity

The theoretical foundation is rooted in classical physics and Lyapunov stability theory, with engineering heritage from the Qian Xuesen hypersonic trajectory paradigm.

2. Axiom System of CDEQ V3.0

The entire framework is based on 16 formal axioms:
13 universal physical/mathematical axioms
3 system boundary axioms

2.1 Thirteen Universal Axioms

1. Energy Conservation (First Law of Thermodynamics)
2. Entropy Increase Principle (Second Law of Thermodynamics)
3. Third Law of Thermodynamics
4. Zeroth Law of Thermodynamics
5. Newton's First Law (Inertia)
6. Newton's Second Law (Force-Motion Relation)
7. Newton's Third Law (Action-Reaction)
8. Conservation of Momentum
9. Conservation of Mass
10. Ideal Gas Law (PV = nRT)
11. Lyapunov Stability Axioms
12. Feedback Control Theory Principles
13. Mathematical Convergence and Limit Axioms

2.2 Three Boundary Axioms (Explicit Constraints)

14. Initial external energy input is required (tank propulsion).
15. Self-sustained thrust comes only from recovered frictional heat.
16. Mode transition requires GCVT validation over 10 consecutive steps.

3. PCTT-CW Pentagonal Cyclic Clockwise Topology (Foundational Core)

中文永久固化版

plaintext
PCTT-CW 五元顺时针循环拓扑定理

最简口诀
五元拓扑，顺时针演化；相邻相济，间隔相制。

核心规则
系统含 5 单元，按顺时针环形演化：
- 相邻：增益辅助
- 间隔：约束制衡

节点与闭环
V={CW0,CW1,CW2,CW3,CW4}
CW0→CW1→CW2→CW3→CW4→CW0

动态公式
A[i,t] = 0.618*P[PREV[i]](t-1) - 0.382*P[PREV2[i]](t-1) + W(t)*ξ[i,t]
P[i,t] = clip(P[i,t-1]+A[i,t], 0.0, 1.0)

系数：S=0.618, C=0.382
索引：i∈{0,1,2,3,4}, PREV[i]=(i-1)mod5, PREV2[i]=(i-2)mod5
初始：P[i,-1]=P[i,-2]=0.5

几何原型（小火人）
人背靠墙站立，头、右手、右腿、左腿、左手五点投影，顺时针连接成环。

最优维度定理
n=3：无抑制，发散崩
n=4：对称死锁，撕裂崩
n=5：生克平衡，稳（唯一最优）
n≥6：冗余内耗，低效崩

证毕。

English Permanent Version

plaintext
PCTT-CW: Pentagonal Cyclic Clockwise Topology

Mantra
Five elements, clockwise evolution. Adjacent reinforce, interval restrain.

Core Rule
5-node clockwise cyclic topology:
- Adjacent: mutual reinforcement
- Interval: mutual restraint

Node Set & Loop
V={CW0,CW1,CW2,CW3,CW4}
CW0→CW1→CW2→CW3→CW4→CW0

Dynamics
A[i,t] = S*P[PREV[i]](t-1) - C*P[PREV2[i]](t-1) + W(t)*ξ[i,t]
P[i,t] = clip(P[i,t-1]+A[i,t], 0.0, 1.0)

Parameters: S=0.618, C=0.382
Index: i∈{0..4}, PREV[i]=(i-1)mod5, PREV2[i]=(i-2)mod5
Initial: P[i,-1]=P[i,-2]=0.5

Geometric Prototype (Little Fire Man)
Person standing back-against-wall: head, R-hand, R-leg, L-leg, L-hand project 5 points, connected clockwise.

Optimality Theorem
n=3: no restraint → divergence (unstable)
n=4: dual deadlock → oscillation (unstable)
n=5: balanced loop → stability (optimal)
n⩾6: redundant overhead → inefficiency

Q.E.D.

4. PCTT-CW Topology and Verilog Implementation

The five-node cyclic topology PCTT-CW is the core dynamical engine of CDEQ. Two complementary implementations realize convergence (V2.0) and self-sustained divergence (V3.0).

4.1 PCTT-CW V2.0 (Convergence)

verilog
```
module pctt_cw();
real P[0:4],A[0:4];
real S=0.618,C=0.382;
integer PREV[0:4]='{4,0,1,2,3};
integer PREV2[0:4]='{3,4,0,1,2};

initial begin
P[0]=0.5;P[1]=0.5;P[2]=0.5;P[3]=0.5;P[4]=0.5;
end
```

```
task update;
integer i;
for(i=0;i<5;i=i+1) begin
A[i]=S*P[PREV[i]]-C*P[PREV2[i]];
P[i]=P[i]+A[i];
P[i]=P[i]>1.0?1.0:P[i]<0.0?0.0:P[i];
end
endtask

endmodule
```

4.2 PCTT-CW V3.0 (Divergence / Self-Sustained)

```
verilog
module pctt_cw_v3();
real P[0:4],A[0:4];
real S=0.382,C=0.618;
integer PREV[0:4]='{4,0,1,2,3};
integer PREV2[0:4]='{3,4,0,1,2};

initial begin
P[0]=0.5;P[1]=0.5;P[2]=0.5;P[3]=0.5;P[4]=0.5;
end

task update;
integer i;
for(i=0;i<5;i=i+1) begin
A[i]=S*P[PREV[i]]-C*P[PREV2[i]];
P[i]=P[i]+A[i];
end
endtask

endmodule
```

The only differences are:

- Coefficient swap: S ↔ C
- Removal of clamping in V3.0 to enable self-sustained growth

5. Master Equations (No Garbled Code · 100% Stable)

V2.0 (Convergence & Static Stability)
$\Psi' = \text{clip}[\varphi\Psi_{-1} - \bar{\varphi}\Psi_{-2} + \omega\xi] + \nabla C$
$\rightarrow \Phi<\Gamma$ & $|E_{11}-5|<\varepsilon$ & $|\Sigma-15|<\varepsilon$ $\forall 10$

V3.0 (Divergence → Self-Sustained Cruise)
$\Psi' = \text{clip}[\varphi\Psi_{-1} - \bar{\varphi}\Psi_{-2} + \omega_\text{pulsed}(v,t)\xi] + \nabla T_\text{friction}(v)$
→ $v \geqslant v_\text{crit}$ & $P_\text{thrust} \geqslant P_\text{drag}$ & 10 consecutive steps
→ Self-sustained cruise state

Where:

- ω_pulsed: staged tank thrust (active below v_crit)
- ∇T_friction: friction heat recovery (active above v_crit)

6. Mode Switching Logic

- $v < v_\text{crit}$: Tank-driven mode
- $v \geqslant v_\text{crit}$: Self-sustained aerothermal mode
- Transition: Confirmed only by GCVT-V3

7. Founder's Claim (Official Copyright & Attribution)

I, Jiang Wenjia, hereby formally claim:

1. I am the sole founder, creator, and original author of the entire CDEQ theoretical system, including CDEQ V1.0, V2.0, V3.0.
2. I am the sole founder of PCTT-CW, CENHE-LEX, and GCVT theorems.
3. All concepts, frameworks, master equations, mode-switching logic, and axiom structures are original intellectual property.
4. CDEQ V3.0 is a continuous evolution of my original stability and propulsion theory.
5. This work is independently completed and published under my sole authorship.

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8. Conclusion

CDEQ V3.0 establishes a new propulsion paradigm: critical velocity mode switching with

aerothermal heat recovery. It is thermodynamically valid, physically consistent, stable, verifiable, and non-perpetual.

The system turns frictional loss into usable thrust, extending the legacy of Qian Xuesen's hypersonic philosophy into long-endurance self-sustained flight.

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